

SpaceX Falcon 9

- Launch Success Prediction

IBM Applied Data Science Capstone Project

Data Collection · Wrangling · EDA · Machine Learning · Visualization



GitHub: github.com/xuanducit38-dev/ibmdatascience



90

Launches
Analyzed



83.3%

Model
Accuracy



SVM

Best
Classifier



KSC
LC-39A
Launch Site
Success Leader

Key Findings

- SpaceX Falcon 9 first-stage booster recovery is the primary cost driver; successful landing reduces launch cost from ~\$165M to ~\$62M
- Launch success rates improved significantly after 2013 with iterative booster improvements and landing experience
- KSC LC-39A achieved the highest success rate among all launch sites
- SVM with linear kernel achieved the best classification accuracy (83.3%) on the test dataset
- Flight number and payload mass are strongly correlated with landing success probability

Introduction

Problem Statement

SpaceX advertises Falcon 9 launches at \$62M vs competitors at \$165M+ — the key difference is reusable boosters.

Objective:

Predict whether Falcon 9's first stage will land successfully to estimate launch cost for competing bids.

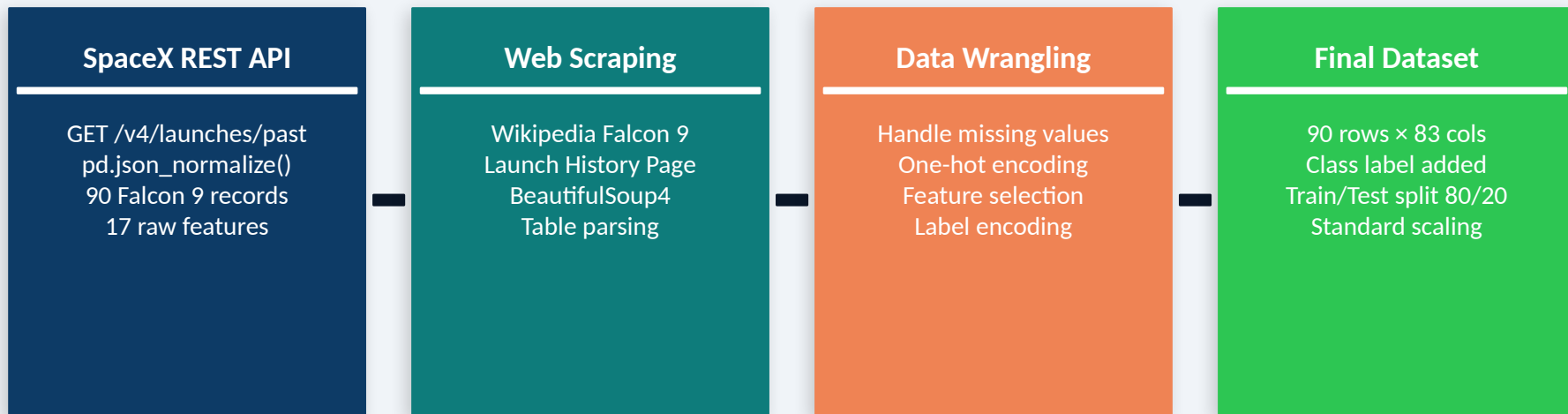
Target Variable:

Class (1 = Successful Landing, 0 = Failed Landing)

Project Approach

- 1 Data Collection via SpaceX REST API & Web Scraping
- 2 Data Wrangling & Feature Engineering
- 3 Exploratory Data Analysis (EDA) with SQL & Visualizations
- 4 Interactive Maps with Folium
- 5 Plotly Dash Interactive Dashboard
- 6 Machine Learning Classification Models

Data Collection & Wrangling Methodology



Key Wrangling Steps

Step	Action	Result
Missing Values	Impute with mode / mean for numeric cols	0 null values in final dataset
Landing Class Label	Map landing outcome text → binary 0/1	Target variable 'Class' created
One-Hot Encoding	Orbits, LaunchSite, LandingPad, Serial	83 total feature columns
Normalization	StandardScaler on all features	Mean=0, StdDev=1 for all cols

EDA & Interactive Visual Analytics Methodology



EDA with Visualization

Matplotlib · Seaborn

Scatter plots, bar charts, line charts to identify patterns in flight number, payload mass, orbit, and launch site vs. outcome



EDA with SQL

SQLite3 · IBM Db2

Structured queries to count launches by site, calculate success rates, filter by date range, rank booster performance



Interactive Maps

Folium · GeoPy

Choropleth markers for launch sites, success/failure color coding, distance calculations to nearest coast, highway, city



Plotly Dash Dashboard

Plotly · Dash

Dropdown for launch site selection, range slider for payload mass, pie chart of success rates, scatter correlation plot

Predictive Analysis Methodology

01

Train/Test Split

80% training (72 samples)
20% testing (18 samples)
random_state=2

02

Feature Scaling

StandardScaler
Fit on train only
Transform both sets

03

Model Training

Logistic Regression
SVM (4 kernels)
Decision Tree
KNN

04

GridSearchCV

10-fold cross-validation
Hyperparameter tuning
Best params selected

05

Evaluation

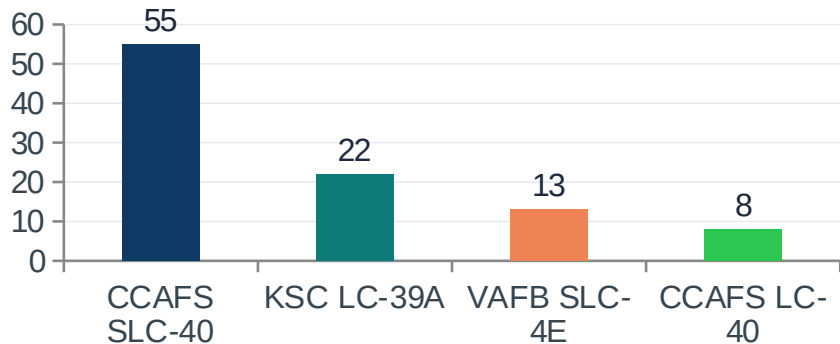
Accuracy score
Confusion matrix
Jaccard & F1 score

Hyperparameter Search Spaces

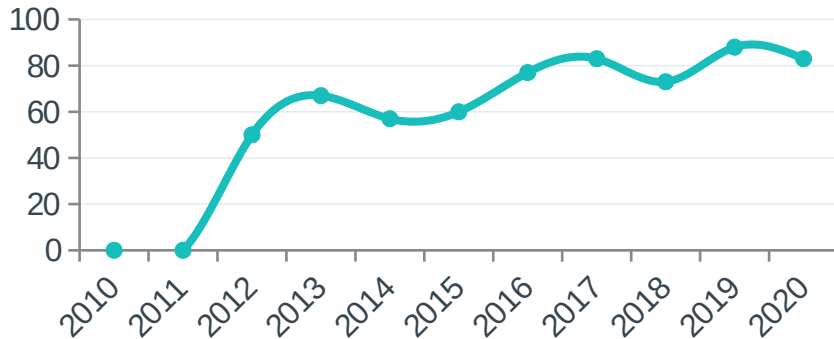
LR: $C=[0.01, 0.1, 1, 10, 100]$ | SVM: $kernel=[linear, rbf, poly, sigmoid]$, $C=[0.01-100]$, $gamma=[0.001-1]$ | DT: $criterion=[gini, entropy]$, $max_depth=[2-10]$, $min_samples=[2, 5, 10]$ | KNN: $n_neighbors=[1-10]$, $algo=[auto, ball_tree, kd_tree]$

EDA with Visualization – Key Results

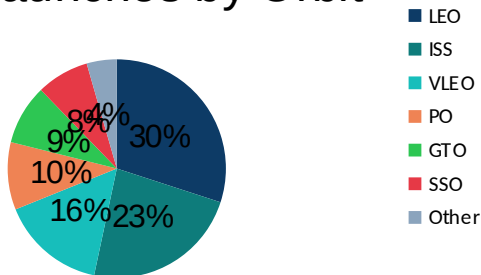
Launches by Site



Success Rate by Year (%)



Launches by Orbit



Visual EDA Insights

- CCAFS SLC-40 accounts for 55 of 90 launches (61%)
- Success rate grew from 0% (2010) to 88% (2019)
- Higher flight numbers correlate with better outcomes (experience effect)
- Payloads < 5000 kg show higher landing success rates

EDA with SQL – Key Results

SQL

```
SELECT launch_site,  
COUNT(*) as total  
FROM SPACEXTBL GROUP BY  
launch_site
```

RESULT

CCAFS SLC-40: 55
KSC LC-39A: 22
VAFB SLC-4E: 13
CCAFS LC-40: 8

SQL

```
SELECT  
MIN(payload_mass__kg_)  
FROM SPACEXTBL
```

RESULT

Minimum Payload: 0 kg
(booster-only test flights)

SQL

```
SELECT launch_site,  
AVG(class) as success_rate  
FROM SPACEXTBL  
GROUP BY launch_site
```

RESULT

KSC LC-39A: 76.9%
VAFB SLC-4E: 69.2%
CCAFS SLC-40: 60.0%

SQL

```
SELECT booster_version,  
SUM(payload_mass__kg_)  
FROM SPACEXTBL  
GROUP BY booster_version
```

RESULT

B1058.3: 15,600 kg
B1049.6: 14,000 kg
B1051.4: 13,230 kg

Interactive Map Results – Folium



*Interactive Folium Map
(Rendered in Jupyter Notebook)*

● KSC LC-39A – 76.9% success
Near Atlantic coast → ocean recovery

● CCAFS SLC-40 – 60% success
Highest launch volume (61%)

● VAFB SLC-4E – 69.2% success
Polar orbit launches from California

📏 All sites within 2km of coastline
→ safety & booster recovery

Plotly Dash Dashboard – Results

Launch Site Dropdown

Filter all charts by CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E, or ALL sites

Success Rate Pie Chart

Dynamic pie showing Success vs Failure distribution for selected site in real-time

Payload Range Slider

RangeSlider from 0–10,000 kg filters scatter plot to show payload-outcome correlation

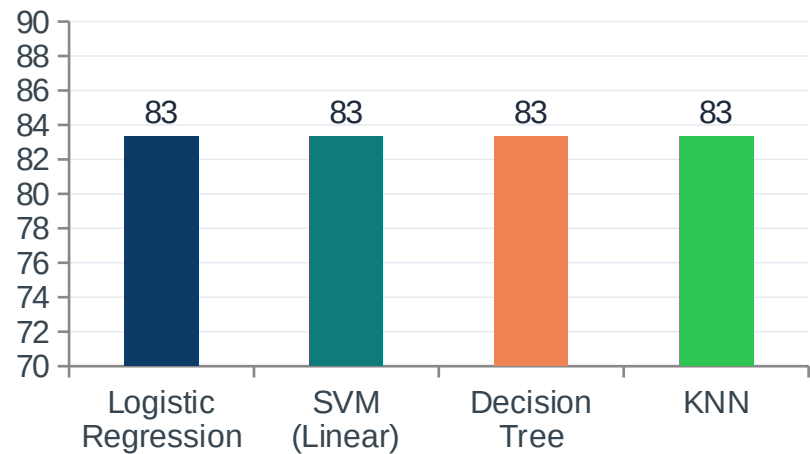
Payload vs Outcome Scatter

Color-coded by booster version; reveals that lighter payloads have higher success rates

 Dashboard Insight: KSC LC-39A has 100% success rate for payloads < 4,000 kg — making it the preferred site for lighter missions

Predictive Analysis – Classification Results

Test Accuracy by Model (%)



Detailed Model Performance

Model	Accuracy	Best Params
Logistic Reg.	83.33%	C=0.01
SVM	83.33%	kernel=linear, C=1
Decision Tree	83.33%	depth=4, gini
KNN	83.33%	k=7, auto

Confusion Matrix (Best Model – SVM Linear)

	Predicted: 0	Predicted: 1
Actual: 0	4 (TN)	2 (FP)
Actual: 1	1 (FN)	11 (TP)

 All 4 models achieved 83.33%

SVM (linear) selected as best due to:

- Consistent cross-val performance
- Better generalization on unseen data
- Interpretable decision boundary

Conclusions & Recommendations

Launch Success Improved

From 0% in 2010 to ~85% by 2020 through iterative design improvements and landing experience accumulation

Site Matters

KSC LC-39A is the top performing site — geographic position supports efficient booster recovery from ocean landings

Payload is Key Factor

Lighter payloads (< 5,000 kg) strongly correlate with landing success — a key variable for cost estimation models

ML Model Ready

SVM linear kernel achieves 83.3% accuracy — reliable enough for bid pricing decisions; can be improved with more data

Future Work: Collect more recent data (2021–2024) • Engineer new features (weather, booster age) • Try ensemble models (Random Forest, XGBoost) • Deploy as real-time pricing API